YZV411E — Big Data Analytics Project Proposal

Flight Delay Propagation and Airport Operational Resilience

Name(s): _____ **ID(s):** _____

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Abstract

Flight delays can affect not only the flight that experiences the initial disruption but also later flights operated by the same aircraft. This project proposes a large-scale analysis of flight delays to investigate where and under which conditions delays are associated with subsequent operational disruptions. Using a multi-year U.S. flight dataset, we will examine temporal and airport-level delay patterns and test whether consecutive flights can be linked using aircraft identifiers and timestamps. Apache Spark will be used to clean, transform, and aggregate the data, and its performance will be compared with a single-machine baseline on representative workloads. The intended output is an evidence-based analysis and a dashboard prototype that may help identify operational patterns relevant to planning.

The dataset, its license, field availability, and the course's minimum 1 GB data requirement will be verified before the final scope is fixed; no causal effect will be claimed from correlation alone.

1. Project Aim and Motivation

- **Problem:** Delays may be associated with later disruptions when aircraft operate multiple flights in sequence. Aggregate delay statistics alone may not reveal these relationships.
- **Research question:** Which airport, time, and flight-sequence patterns are associated with delay propagation, and how can these patterns inform operational planning?
- **Practical value:** Findings may help analysts identify periods or locations where schedule buffers and operational monitoring deserve closer attention.
- **Big Data motivation:** Multi-year flight records require substantial cleaning, grouping, joins, and time-based analysis. Distributed processing can support scalable processing and systematic runtime comparisons.

The project will be exploratory. Relationships between delays will be interpreted as associations unless a suitable causal identification strategy can be justified.

2. Data

Candidate dataset: Flight Delay Dataset 2018–2022, available through Kaggle:

https://www.kaggle.com/datasets/robikscube/flight-delay-dataset-20182022

The dataset is based on U.S. flight records. Candidate fields include flight date, airline, aircraft tail number, origin and destination airports, scheduled and actual departure/arrival times, departure/arrival delay, cancellation, diversion, and delay-cause fields. Exact field names and availability will be confirmed from the downloaded files and documentation.

- **Type and format:** Structured tabular data, expected to be distributed as CSV files.
- **Scale requirement:** The course requires at least 1 GB of data. We will measure actual local file sizes and record the selected years and total volume. If the chosen subset is below 1 GB, additional years or appropriate files will be included, subject to availability and license.
- **Data quality:** Inspect schema, duplicate records, missing values, timestamp consistency, delay distributions, and cancellation/diversion encoding.
- **Sequence linking:** If aircraft identifiers and timestamps are sufficiently complete, sort flights by aircraft and time to identify plausible consecutive-flight pairs. Ambiguous matches and missing identifiers will be handled explicitly.
- **Partitioning:** Preserve raw data; partition processed data by year and, where useful, month. Compare partitioning choices for time-window and airport-level queries. Convert cleaned data to Parquet.

- **Access and license:** Confirm current download requirements, dataset documentation, and permitted academic use.

3. Tools and Frameworks

- **Apache Spark (PySpark):** Distributed cleaning, transformations, joins, aggregations, and flight-sequence analysis.
- **Python and Pandas:** Schema inspection, small-sample validation, and a single-machine baseline where the workload fits in memory.
- **Parquet:** Columnar storage for cleaned data and repeated analytical queries.
- **Matplotlib/Plotly:** Visualize delay distributions, airport/time patterns, and performance results.
- **Jupyter Notebook:** Reproducible experiments and documentation.

Kafka and streaming infrastructure are not planned for the initial scope because the candidate dataset is historical rather than a live stream.

4. Methodology

1. Download candidate files and document file sizes, schema, source, and license.
2. Clean and validate timestamps, identifiers, duplicates, missing values, and delay fields.
3. Produce baseline descriptive analyses by airport, airline, hour, weekday, and season.
4. Where supported, link consecutive flights by aircraft identifier and chronological order; quantify how one flight's delay relates to the next flight's departure delay.
5. Compare selected time periods and examine whether unusual periods, including the COVID-19 disruption, change the patterns.
6. Run representative workloads using a single-machine baseline and Spark. Record runtime, data volume, partition settings, and speed-up where comparison is fair.
7. Translate robust findings into cautious operational recommendations and a dashboard prototype.

5. Expected Outcomes and Evaluation

- A documented, reproducible data-processing pipeline.
- A descriptive report on delay patterns across airports and time periods.
- An assessment of whether aircraft and timestamp fields support reliable consecutive-flight analysis.
- An analysis of associations between an earlier flight's delay and a subsequent flight's delay, with limitations clearly stated.
- A small dashboard or set of visualizations for operational insights.
- A performance comparison between single-machine processing and Spark.

Evaluation will use data-quality statistics, coverage of linked flight sequences, sensitivity checks across time periods, and runtime measurements on the same representative workloads. If predictive modelling is later added, temporal train/test separation will be used to reduce leakage. The project will not claim that an observed association proves causation or that recommendations reduce delays without further validation.

6. Progress Plan

1. **Dataset validation:** Verify download access, documentation, license, fields, and the 1 GB minimum.
2. **Literature review:** Review research on flight-delay propagation, airport resilience, and scalable flight-data processing; define the research gap.
3. **Data preparation:** Build schema and data-quality checks; create cleaned Parquet data.
4. **Exploratory analysis:** Establish airport- and time-level delay baselines.

5 **Propagation analysis:** Test aircraft-based sequence linking and quantify associations if data quality permits.

6 **Big Data experiments:** Compare Spark and single-machine performance and test partitioning choices.

7 **Final outputs:** Prepare visualizations/dashboard, limitations, recommendations, and final report.

Preliminary References

- Kaggle, *Flight Delay Dataset 2018–2022*: dataset page.
- U.S. Bureau of Transportation Statistics, TranStats: https://www.transtats.bts.gov/.